UNIVERSITY OF BUEA

FACULTY OF ENGINEERING AND TECHNOLOGY

CEF440

INTERNET PROGRAMMING AND MOBILE PROGRAMMING GROUP 16

TASK 3: REQUIREMENT ANALYSIS

Submitted to:

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I. REVIEWING AND ANALYSING THE REQUIREMENTS GATHERED

1.1.Introduction

Reviewing and analysing gathered requirements is a critical, iterative phase in software development designed to ensure that the raw stakeholder needs are transformed into a clear, actionable, and consistent blueprint for development.

The objective of this phase is to critically examine the gathered requirements for completeness, clarity, technical feasibility, and dependency relationships, while identifying inconsistencies, ambiguities, and missing information before proceeding to the system design phase.

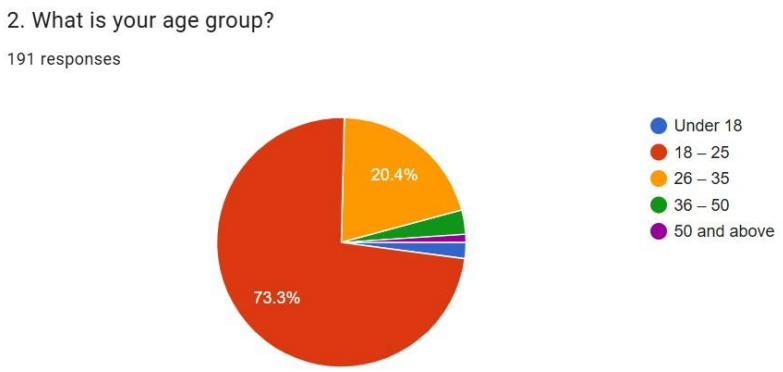
This report presents the findings of the Requirements Analysis phase for our project titled **"Design and Implementation of a Mobile App for Collection of User Experience Data from Mobile Network Subscribers."** The analysis is based on structured survey data collected from 190 mobile network subscribers across multiple cities in Cameroon, including Buea, Yaounde, Douala, Bamenda, Limbe, Dschang, Kumba, and Tiko.

1.2 Survey Overview and Respondent Profile

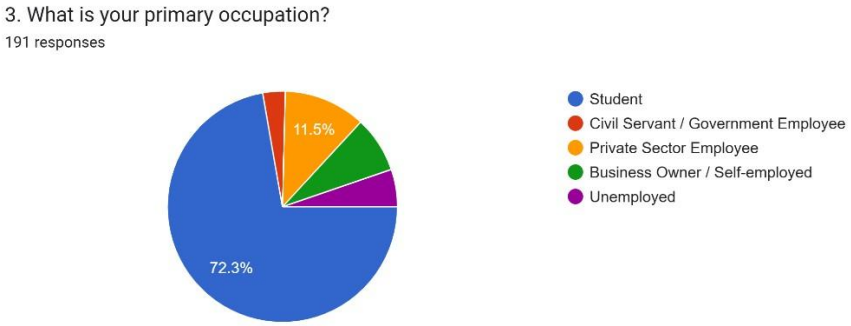
1.2.1 Respondent Demographics

Our survey gathered responses from 190 participants, providing a substantial and representative base for requirements elicitation. A demographic breakdown of our respoondents is summarized below.

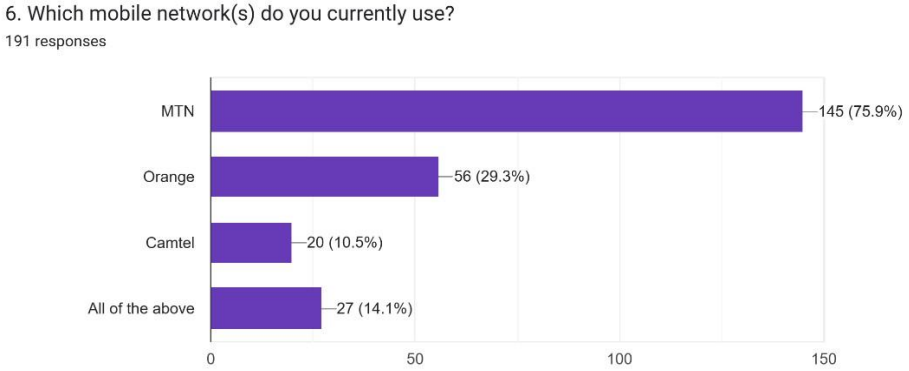
- Age Group: 18 – 25years being the dominant age group giving a 73.9% (139/190) dominance for that age group.



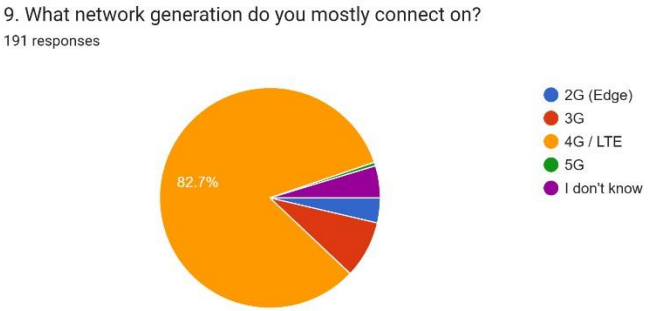
- Occupation: Majority of the respondents were Students with a dominance percent of 72.9% (137/190)



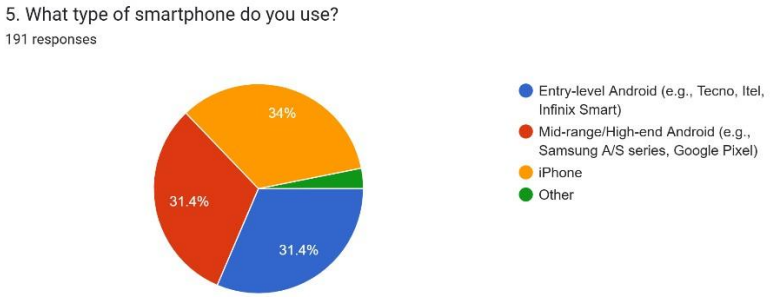
- Primary Network: MTN was dominant as the most used network over Orange and Camtel with 143 respondents selecting it as their main network.



- Network Generation: Majority of the users use the 4G/LTE network



- Smartphone Type: Most the respondents use Mid-range/High-end Android (e.g.,Samsung A/S series, Google Pixel)

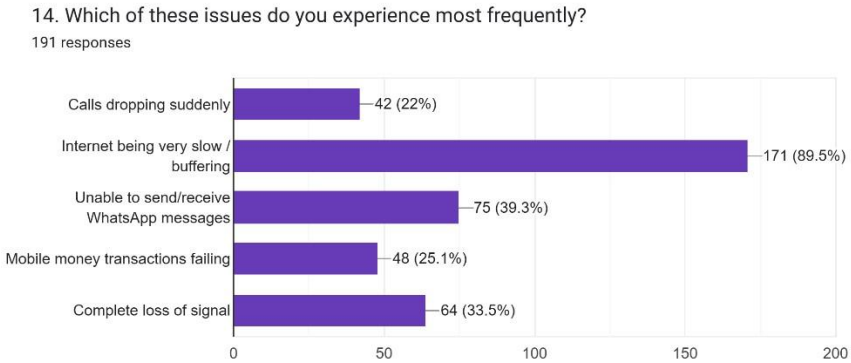


The survey was heavily filled by young adults (18–25) and students, which is significant context for the app design. Meaning our app must be intuitive, resource-efficient, and incentive-driven to appeal to this demographic of our users.

1.3 Completeness Analysis

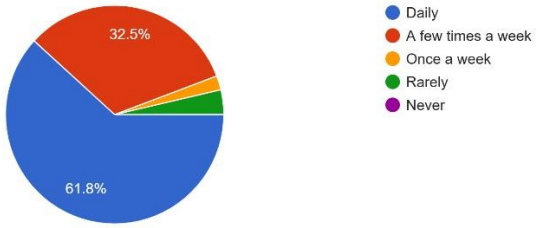
Our survey successfully brought information across several key requirement areas critical to our application:

- User Pain Points: Poor internet speed/buffering was identified by 168 of 190 respondents (89.1%) as the most frequent issue, followed by WhatsApp messaging failures (75), complete signal loss (63), mobile money transaction failures (47), and dropped calls (41). These provide a clear empirical basis for the app's monitoring scope. The analysis is shown below;



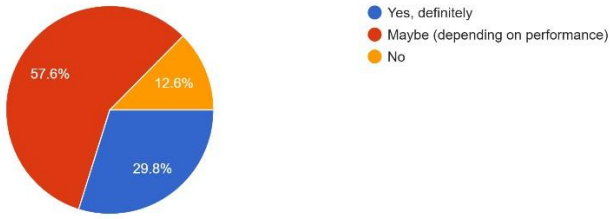
- Usage Patterns: 116 respondents (61.7%) experience poor network quality daily, and a further 61 (32.4%) experience it several times a week, confirming the sustained need for continuous background monitoring. As seen below;

13. How often do you experience poor network quality (slow speed, dropped calls, no signal)?
191 responses



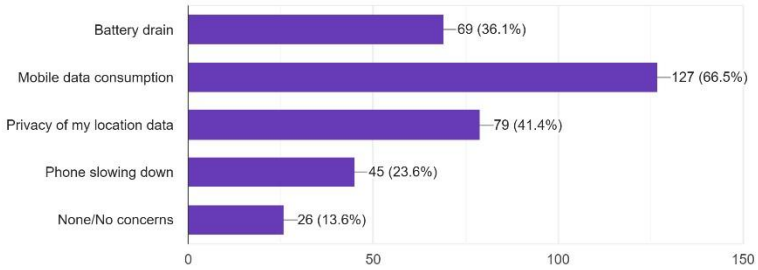
- App Adoption Willingness: 56 respondents (29.8%) would definitely install the monitoring app; 109 (58.0%) said "maybe depending on performance"; and only 23 (12.2%) refused outright. This gives a conditional adoption rate of up to 87.8%, supporting the project's viability.

19. Would you install an app that quietly monitors your network quality in the background for research purposes?
191 responses

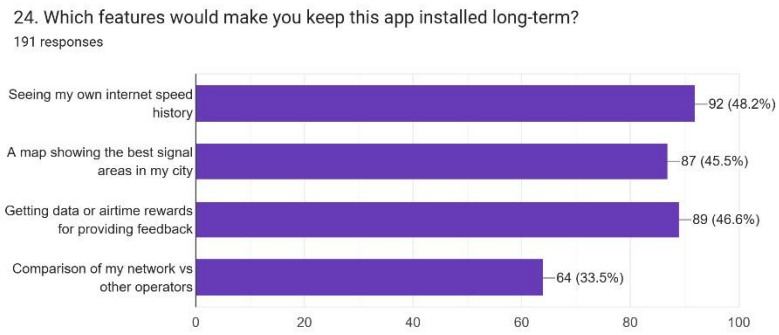


User Concerns (Non-Functional Requirements): The top concerns about the app mobile data consumption (stated in 134 responses across combinations), location privacy (about 100 responses), battery drain (about 90 responses), and phone performance (about 50 responses) directly translate into non-functional requirements around efficiency, privacy, and minimal resource footprint.

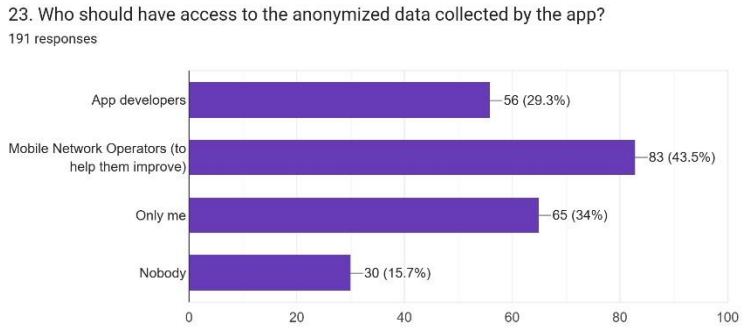
20. What would be your main concern about using such an app?
191 responses



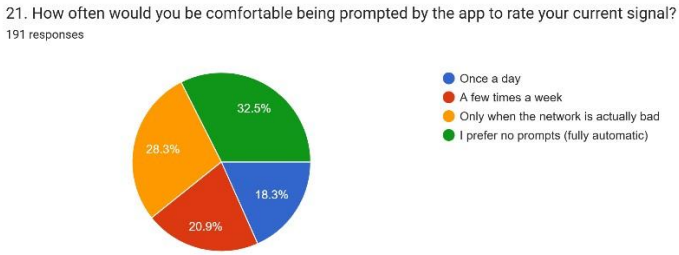
- Desired Features: Respondents clearly ranked their preferred features: personal internet speed history (91), data/airtime rewards (88), a signal quality map (86), and operator comparisons (63). These formed a well-defined feature backlog.



- Data Governance Preferences: The majority prefer that anonymized data be shared only with Mobile Network Operators (82) or themselves (65), followed by app developers (55), with 29 preferring no access at all. This means the want their data to reach the operators so their issues can be resolved.



- Notification Preferences: 62 respondents prefer fully automatic (silent) monitoring with no prompts, 53 want prompts only when the network is bad, 38 want prompts a few times a week, and 35 are comfortable with once-daily prompts. This defines the push notification strategy.



1.4 Clarity Analysis

The following requirements are stated with sufficient clarity to proceed to design:

- The app must monitor network quality passively in the background without requiring active user participation at all times.

- The app must allow users to view their own historical network speed data.
- The app must display a geographic map indicating signal quality across locations.
- The app must support a comparison view between different mobile network operators.
- The app must be configurable with respect to how frequently it prompts the user for manual signal ratings.
- GPS location collection must be optional or limited to approximate location, with explicit user consent.
- Data collected must be anonymized before being shared with any third party.

1.5 Technical Feasibility

1.5.1 Feasible Requirements

The following required capabilities are technically achievable within standard mobile development frameworks (Android/iOS native or cross-platform such as Flutter or React Native):

- **Background Network Monitoring:** Reading signal strength (RSSI/RSRP), network type (2G/3G/4G/5G), and data speeds is supported via Android's TelephonyManager API and iOS's CoreTelephony framework, with appropriate permissions.
- **GPS-tagged Readings:** Passive GPS logging alongside network reads is standard practice in drive-test and crowdsourcing apps. Approximate location can be achieved by rounding coordinates to a defined grid.
- **Speed History Dashboard:** Storing timestamped metric readings locally (SQLite/Room database) and presenting them in a chart view is well within standard development scope.
- **Signal Coverage Map:** Rendering user-contributed readings as a heatmap or marker overlay using Google Maps SDK or OpenStreetMap is technically feasible and widely demonstrated in tools such as OpenSignal.

- **Push Notification / In-App Prompts:** Firebase Cloud Messaging (FCM) or local notifications can implement configurable user prompts; respecting Do Not Disturb settings.
- **Data Upload to Backend:** REST API-based transmission of anonymised readings to a cloud backend (e.g., Firebase, AWS, or a custom server) is standard and welldocumented.

1.5.2 Technical Challenging Requirements

The following requirements require more onformation to confirm feasibility within our project's constraints:

- 5G Monitoring: One respondent reported using 5G, but commercial 5G deployment in Cameroon is limited. The app should be designed to accommodate future 5G metric sets without requiring a full rewrite.
- Operator Comparison (Cross-Operator Data): Real-time comparison between MTN, Orange, Camtel, and other operators requires either dual-SIM API access (available on some Android devices but complex) or crowdsourced data aggregation from multiple users' readings. Crowdsourcing introduces minimum viable population requirements before the map becomes meaningful.
- Data Anonymization at Scale: Providing genuinely strong anonymization requires implementing privacy-preserving techniques such as differential privacy or spatial k-anonymity, which add significant backend complexity.
- iOS Background Restrictions: Apple imposes strict limits on background processing for battery conservation. Continuous background network monitoring on iOS is significantly more restricted than on Android. This may require the app to be iOS-exclusive in a limited form, or rely on periodic silent push notifications to trigger readings.

1.6 Dependency Relationships

The dependency table below identifies the relationships between key system components and requirements and also brings out the dependency type. User Consent → OS Permissions → Background Monitoring → Local Storage → Data Upload → Coverage Map.

Feature / Requirement	Depends On	Dependency Type
Background Network Monitoring	OS permissions (location & network state)	Prerequisite
GPS Signal Readings	Background Network Monitoring & GPS permission	Sequential
Local Data Storage	Background Network Monitoring	Sequential
Signal Coverage Map	GPS-Tagged Readings & Crowdsourced Backend	Sequential & External
Internet Speed History	Local Data Storage & UI charting component	Sequential
Operator Comparison View	Crowdsourced Backend & Multi-user Data Volume	Sequential & Scale
Manual Signal Rating (Prompt)	Push notification system & User preferences store	Parallel (independent)
Data Upload to Cloud	Anonymization module & Backend API & Network connection	Sequential
User Consent & Privacy Settings	All data collection features	Foundational prerequisite

1.7 Inconsistencies Identified

The following inconsistencies were observed in our survey data which has to be resolved, as they reflect conflicting user expectations that cannot all be simultaneously satisfied without trade-offs:

Inconsistency	Detail	Implication
Privacy vs. Location Mapping	47 respondents said 'No' to GPS logging, yet 86 want a signal coverage map, which requires location data to generate.	The map feature cannot be delivered without GPS data. Consent and education about approximate location must be built into onboarding.

No Prompts vs. Feedback Rewards	62 respondents prefer fully automatic (no prompts) monitoring, yet 88 want rewards for providing feedback which typically implies active participation.	A reward mechanism for passive-only users needs a different model (e.g., time-based or reading-countbased) rather than perprompt rewards.
Data Privacy vs. MNO Data Sharing	Some respondents want only they themselves to access data (65), while 82 support MNO access. These groups overlap in unpredictable ways.	The consent model must allow granular, per-user choices rather than a single global policy.
Satisfaction Paradox	9 respondents rated satisfaction as 5/5 yet also reported experiencing daily poor quality.	Satisfaction scores may reflect relative expectations rather than absolute quality. The app should measure objective metrics, not rely on user satisfaction ratings alone.
No Complaint meaning No Problem	56 respondents said they didn't complain because they 'didn't think it would make a difference'; 70 didn't know	The app's feedback submission feature addresses a real gap in the complaint pathway; this must be framed
	how. Yet 168 of 188 experience slow internet.	clearly in the app's value proposition.

1.7 Ambiguities and Missing Information

The table below shows all identified issues(ambiguity and missing info) from the analysis for stakeholder review and how the issues can be resloved before our design begins.

Issue Type	Description	Action Required
Missing Info	No Android/iOS version distribution data collected	Conduct targeted follow-up or device audit

Missing Info	No defined acceptable battery/data overhead thresholds	Define measurable non-functional requirements
Ambiguity	"Approximate" GPS location undefined in quantitative terms	Specify precision level (e.g., ±500m grid)
Ambiguity	"Anonymized" data lacks technical specification	Define anonymization method (pseudonymisation, k-anon, etc.)
Missing Info	Reward redemption model and MNO integration undefined	Scope reward system; confirm MNO API availability
Ambiguity	Operator comparison relies on undefined data sourcing model	Decide between crowdsourced vs. dual-SIM approach
Inconsistency	GPS opt-out conflicts with coverage map feature	Design map to function with partial location consent
Inconsistency	No-prompt preference conflicts with feedback reward model	Design passive reward accumulation mechanism
Missing Info	Offline storage and sync behaviour not specified	Define local buffering policy and sync frequency
Technical Risk	iOS background processing limitations not addressed	Prototype iOS background capability; scope accordingly
Missing Info	Specific KPIs for network quality (ping, jitter, etc.) not prioritised by users	Present metric options to sample users for prioritisation
Missing Info	Regulatory/legal consent framework for passive data collection not addressed	Legal review required before data collection begins

Some other missing information we identified in our survey are;

- Specific Network Metrics of Interest: Respondents described subjective experiences ("slow", "buffering") but our survey did not ask which technical metrics (download speed, upload speed, ping/latency, signal strength in dBm, jitter, packet loss) they would find most meaningful to see in the app.

- Acceptable Battery and Data Overhead: Users flagged battery drain and data consumption as concerns, but no threshold was established (e.g., "less than 2% battery per hour" or "less than 10MB/day"). These thresholds are required as testable non-functional requirements.
- Background App Permission Acceptance: No question directly assessed willingness to grant background location and network access permissions, a critical requirement for passive monitoring on modern mobile OS.
- Offline Functionality Expectation: Given that some respondents experience complete network outages, it is unclear whether the app should store readings locally and sync later, or discard readings when offline.

II. PRIORITIZED REQUIREMENTS

Based on the analysis of the 190 survey responses, we have categorized the user requirements into four priority groups using an **Importance and Feasibility Matrix**.

2.1 High Importance / High Feasibility

These are "Quick Wins" that users value most and are technically implemented using standard mobile APIs.

Requirement	Importance	Feasibility	Rationale
Personal Speed History	Highest (92 mentions)	High	Users want to track their own experience. This only requires local storage and simple logging.
Network Coverage Maps	High (87 mentions)	High	Aggregating signal strength data into a heatmap is a standard GIS implementation.
Passive Monitoring	High (62+ votes)	High	Majority prefer "Fully Automatic" or "Only when bad" prompts. Passive background collection is highly feasible.
Network Comparison	Moderate (64 mentions)	High	Comparing performance between operators (MTN vs. Orange vs. Nexttel) using anonymized crowd data

2.2 High Importance / Medium Feasibility

These are critical to user retention but require more complex backend development, optimization, or external partnerships.

Requirement	Importance	Feasibility	Rationale
Low Data Consumption	Crucial (127 concerns)	Medium	The number one concern. Optimizing data packets so the app doesn't consume the user's expensive mobile data is vital but technically demanding.
Low Battery Drain	High (69 concerns)	Medium	Background GPS and signal scanning are battery-intensive. Requires sophisticated scheduling and sensor batching.

2.3 Moderate Importance / High Feasibility

These add value and address specific user concerns without requiring major investment.

Requirement	Importance	Feasibility	Rationale
Approximate Privacy	High (85 votes)	High	Users are more comfortable with "Approximate Location" than exact GPS. This is a simple settings-level configuration.
Performance Optimization	Moderate (45 concerns)	High	Concerns about "Phone slowing down" can be addressed through standard lightweight coding practices and small APK sizes.

2.4 Low Importance / Low Feasibility

These are features or requirements that had low interest or would be too difficult to implement effectively.

Requirement	Importance	Feasibility	Rationale
Manual Frequent Prompts	Low	High	Only 35 users wanted daily prompts. Most find it annoying, so this should not be a primary feature.
Exact Real-time GPS	Low	Medium	63 users explicitly said "No" to GPS tracking. Implementation is easy, but it creates a massive barrier to user trust.

Summary Recommendation

To ensure the app is successful and used by the users we will ensure the following requirement

- **Must-Haves:** we will focus on the **Speed History** and **Coverage Maps** first.
- **User Trust:** Address the **Data Consumption** concern immediately by including a "Data Saver" mode or showing the user exactly how many KBs the app has used.

2.5 Critical Importance (Must-Haves)

These requirements address the most cited pain points and the primary concerns that would prevent users from installing or keeping the app.

- **Minimal Data Consumption:** This is the first priority of the users. Because of how data is expensive users will un install any app thaat consumes data
- **Minimal Battery Impact:** These is one of the High concern among users. The app must run in the background without noticeably draining the smartphone's power.
- **Privacy & Approximate Location:** A significant portion of users (nearly 1/3) are uncomfortable with GPS tracking. Providing "Approximate Location" or high-level privacy controls is essential for user trust.
- **Reliability for "No Signal" Scenarios:** Since "Daily" network issues and "Complete loss of signal" were frequently reported, the app must be able to log data offline and sync when the network returns.

2.6 High Importance (Should-Haves)

These are the features users explicitly asked for as incentives to keep the app on their phones.

- **Personal Speed History:** The most requested feature (92 votes). Users want to see a log of their own internet performance over time.
- **Coverage/Signal Maps:** Very high interest (87 votes). Users want to see which areas in their city have the best signal before they travel or visit.

2.7 Moderate Importance (Could-Haves)

These features provide competitive value and additional insights but are not the primary drivers for installation.

- **Operator Comparison:** Users are interested in seeing how their network (e.g., MTN) performs compared to others (e.g., Orange) in specific locations.
- **Automated Background Monitoring:** Most users prefer the app to work quietly in the background without constant prompts, making "full automation" a preferred UX requirement.
- **Formal Complaint Support:** Since many users mentioned they don't complain because they "don't know how" or "don't think it helps," a feature that helps generate report data for operators would be valuable.

2.8 Low Importance (Won't-Haves for now)

These requirements had low user interest or were actively viewed as negatives.

- **Frequent Manual Rating Prompts:** Only a small minority of users want to be prompted daily to rate their signal.
- **Real-time Exact GPS Tracking:** Due to privacy concerns, exact coordinates are less important to users than general location-based Based on the technical effort, resource requirements, and implementation complexity for a mobile application, the requirements are classified below by **Feasibility**.

2.9 High Feasibility (Easiest to Implement)

These requirements rely on standard Android/iOS APIs and local data processing, making them straightforward to build with minimal external dependencies.

- **Personal Internet Speed History:** This involves logging local test results to a simple on-device database (like SQLite or Room) and displaying them via a list or chart. It requires no external data sources other than the app's own logs.
- **Approximate Location Tracking:** Standard mobile OS permissions allow for "Coarse Location" (WiFi/Cell tower triangulation) instead of "Fine Location" (GPS). This is easier on the battery and requires less precise data processing.
- **Passive Background Monitoring:** Modern mobile frameworks provide "Work Managers" or "Background Tasks" specifically designed to perform periodic network pings or signal strength checks without active user interaction.
- **Basic Complaint Generation:**
    - **Reasoning:** This is a UI-driven feature that packages collected network logs into a formatted text or PDF file that the user can share via email or WhatsApp.

2.10 Medium Feasibility (Moderate Effort)

These requirements are technically possible but require significant optimization, backend infrastructure, or careful handling of OS restrictions.

- **Network Coverage & Signal Maps:** While displaying a map is easy, this requires a robust backend to aggregate data from hundreds of users, clean the data points, and generate a heatmap that is served back to the app in real-time.
- **Low Data/Battery Consumption Optimization:** Achieving "zero-footprint" monitoring is difficult. It requires advanced knowledge of background execution limits (especially on iOS) and "batching" data uploads to ensure the app doesn't trigger high power usage or consume significant data.
- **Operator Comparison (Benchmarking):** This requires a large enough sample size of data from all major operators (MTN, Orange, Nexttel, Camtel) to be statistically accurate and useful for the user.

2.11 Low Feasibility (Most Challenging)

These requirements involve complex external factors, high operational costs, or legal/security hurdles.

- **Cross-Operator Real-Time Switching:** While users may want to know which SIM is better at a given moment, mobile operating systems have strict security limits that prevent third-party apps from automatically switching data or voice traffic between SIM cards.
- **Highly Frequent Manual Prompts:** While easy to code, OS "Do Not Disturb" settings and background execution policies often kill apps that try to push too many notifications, making it technically unreliable to hit a specific daily frequency.

III. REQUIREMENT CLASSIFICATION

3.1 Functional Requirements

These are specific behaviors and features a system must perform to meet user needs. The following are the identified Functional requirements for the system.

- **Background Network Monitoring**
  The app shall continuously monitor and collect network metrics in the background without requiring the user to open it.

- **Network Metric Collection**
  The app shall automatically collect and log the following metrics with each reading:
  I. Network type
  II. Signal strength
  III. Internet speed(download/upload)
  IV. GPS location
  V. Timestamp

- **User Feedback Prompts**
  The app shall periodically prompt the user to rate their current network experience.

- **Fully automatic Mode**
  The app shall offer a fully automatic mode where no prompts are shown and all data is collected silently

- **Internet speed history**
  The app shall store and display a personal history of the user's internet speed over time.

- **Network coverage map**
  The app shall display a map showing network signal quality across different location based on collected data.

- **Operator comparison**
  The app shall allow comparison of network performance between different operators.

- **Anonymous Data sharing**
  The app shall allow anonymized collected data to be shared with mobile network operators to support network improvement.

- **Offline data storage**
  The app shall store all collected data locally on the device when there is no internet connection and sync when connectivity is restored.

- **Network problem alerts**
  The app shall notify the user when an unusual or significant drop in network quality is detected.

3.2 Non-functional requirement

These defines quality standards, how the system must perform.

- **Battery efficiency**
  The app shall consume no more than 5% of device battery per hour during
- **Minimal Data consumption**
  The app shall use less than 10MB of mobile data per day for its own background operation
- **Location Privacy**
  The app shall anonymize all GPS coordinates before storing or transmitting them. Exact location shall never be stored or shared.
- **Android compatibility**
  The app shall be compatible with Android version 8.0 and above, including entrylevel devices such as: Tecno, Itel, and Infinix.
- **Performance on low-end devices**
  The app shall operate without causing noticeable slowdown on entry-level android devices.
- **Usability**
  A first-time user shall be able to navigate the app and understand its main features without any training or manual.
- **Data security**

All data transmitted from the app to any server shall be encrypted to protect user privacy.

- **Reliability**
  The app's background service shall restart automatically if it is stopped by the OS
- **Transparency**
  The app shall clearly inform users what data is being collected, how it is used, and who can access it before they install or use it.
- **Bilingual support**
  The app shall support both English and French languages.

IV. REQUIREMENT VALIDATION WITH STAKEHOLDERS

4.1 Overview

The goal of this phase is to ensure that the requirements we identified are accurate and solve the problems of our primary stakeholders. While we are still awaiting technical feedback from ISP personnel, we have performed a thorough validation using the responses gathered from our Mobile Network Subscriber survey to ensure the project meets user needs.

4.2 Validation Process (User-Centric Focus)

Because our primary goal is to close the gap in "Quality of Experience," we used the data from the subscriber Google Forms to verify our requirements:

Prioritizing User Needs

- We compared our proposed feature list against the specific "fluctuations" and "service delays" reported by users in the survey.
- Subscribers highlighted that unstable connections and high data costs are their biggest hurdles.
- This validated our requirement for the app to collect objective metrics (like latency and packet loss) in the background so we can pinpoint exactly when and where these issues occur.

Verifying Feature Usefulness

- We reviewed user comments regarding "automated and user-friendly collection" of data.

- Users confirmed that a simple satisfaction rating system (overall satisfaction, response time, and usability) is a feature they would actually engage with.
- This validated our requirement to allow users to rate their experience on various dimensions with corresponding timestamps and location data.

4.3 Addressing Missing Information and Constraints

Since we didn't receive responses from ISP personnel, we had to adjust our requirements to be more robust:

- **Dependency Analysis:** We identified a "missing information" risk regarding backend infrastructure limits. To handle this, we added a requirement for the app to be fully autonomous in its logging so it doesn't rely on ISP-side data.
- **User Reluctance:** The survey data showed that users are hesitant about apps that consume too much data. We updated our requirements to include a "Data-Only-on-Wi-Fi" sync option to respect these cost concerns.

By validating our requirements against the available subscriber data, we have ensured our **Software Requirement Specification** (SRS) is:

Clear: The requirements directly address the "incomplete understanding of service quality" reported by users.

Prioritized: We are focusing on features that provide the most value to subscribers in lowresource environments.

Feasible: The metrics we plan to collect (signal strength, network type, etc.) are standard device parameters that do not require ISP intervention to gather.